

Literature Status: An $\mathcal{L}_\infty$ HI Space With No Reflexive Subspace

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Status

This is a lightweight `literature_already_answered` packet. It records that a future-work existence problem announced in Argyros–Motakis, *A dual method of constructing hereditarily indecomposable Banach spaces*, arXiv:1504.01564, is answered in the separate later paper Argyros–Motakis, *The scalar-plus-compact property in spaces without reflexive subspaces*, arXiv:1608.01962.

Source Question

In the introduction of arXiv:1504.01564, PDF page 2, parsed source lines 159–161, the authors discuss the problem of constructing a norming set whose space has a shrinking Schauder basis and no subspace with a boundedly complete basis. They state that this is directly related to the existence of an $\mathcal{L}_\infty$ space that is hereditarily indecomposable and has no reflexive subspace. The paper answers the first problem and announces the second as forthcoming: there exists an $\mathcal{L}_\infty$ HI space with no reflexive subspace.

Supporting Answer

The supporting paper arXiv:1608.01962 cites arXiv:1504.01564 as the dual method paper. Its abstract constructs a hereditarily indecomposable $\mathcal{L}_\infty$ space $\mathfrak{X}_{\text{nr}}$ which contains no c_0 , no ℓ_1 , and no reflexive subspace, and which satisfies the scalar-plus-compact property.

The theorem-level evidence is explicit. Theorem 6.12 states that $\mathfrak{X}_{\text{nr}}$ contains no reflexive subspace. Theorem 6.10 states that every strictly singular operator on $\mathfrak{X}_{\text{nr}}$ is compact, hence $\mathfrak{X}_{\text{nr}}$ satisfies the scalar-plus-compact property. The introduction also states that this is the first example of an $\mathcal{L}_\infty$ HI space not containing a reflexive subspace.

Scope

This is not a new proof. It records the already-known later answer to the $\mathcal{L}_\infty$ HI/no-reflexive-subspace existence statement announced in arXiv:1504.01564. The first norming-set problem in the same source paper is answered by that source paper itself and is therefore not counted as a separate literature-answer packet.

References

- S. A. Argyros and P. Motakis, *A dual method of constructing hereditarily indecomposable Banach spaces*, arXiv:1504.01564.
- S. A. Argyros and P. Motakis, *The scalar-plus-compact property in spaces without reflexive subspaces*, arXiv:1608.01962.